

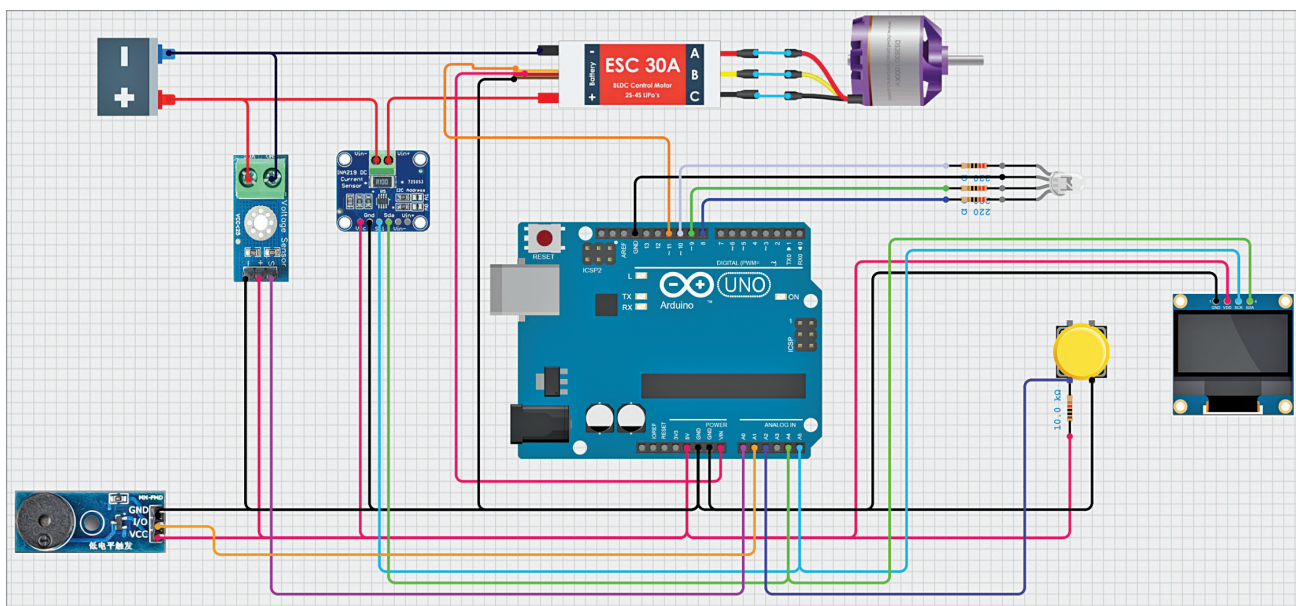


Fig. 5: Wiring diagram of all the components with the Arduino Nano expansion board connections

**TABLE 2**  
**RGB LED STATUS INDICATION**

| RGB Colour         | Status Meaning   | Description                                                    |
|--------------------|------------------|----------------------------------------------------------------|
| Cyan (blue+green)  | Idle/ready       | The system is powered on and ready for testing                 |
| Red (solid)        | Testing/no power | Motor test is running or no battery                            |
| Red (blinking)     | Low power        | Battery voltage is low, unsafe to start the motor              |
| Yellow (red+green) | Error/fault      | Fault detected (ESC, motor, propeller, overload, wiring issue) |
| Green              | Success          | Motor, ESC, and power system are working correctly             |

corresponding RGB status indications are listed in Table 2.

## Circuit and working

Fig. 3 shows the circuit diagram of the BLDC motor failure detection system for drone safety. The system is built around an Arduino Nano, an INA219 current sensor module, an electronic speed controller (ESC), a BLDC motor, a 2.4cm (0.96-inch) I<sup>2</sup>C OLED display, an RGB LED, a voltage sensor module, and several supporting components.

The voltage sensor provides an analogue output that connects to one of the Arduino's analogue input pins. The INA219 current sensor and the OLED display communicate with the Arduino through the I<sup>2</sup>C pins (SDA and SCL).

The 12V LiPo battery used for the BLDC motor connects to the 12V (red) and GND terminals of the ESC.

Additional ESC units, if required, can be connected in the same manner. The ground of the LiPo battery connects to the ground of the voltage sensor module, and the battery's positive terminal connects to the positive input of the voltage sensor.

The BLDC motor connects to the ESC through its three-phase wires. All components follow the circuit diagram shown in Fig. 3, and a prototype was developed as shown in Fig. 1.

To connect the Arduino Nano with an expansion board, first align the Nano carefully with the female headers on the board. Ensure that the USB port faces outward in the correct direction as marked on the expansion board. Match the pins properly (VIN, 5V, GND, D2-D13, A0-A7) before inserting, then press the Nano gently and evenly into place to avoid bending the pins.

Once mounted, the board can be powered via the Nano's USB port, the DC jack on the expansion board (7-12V input), or by supplying a regulated 5V directly to the 5V pin. The ground (GND) must remain common for all connected devices.

The expansion board simplifies wiring by providing three pins for each I/O: Signal (S), VCC (5V), and Ground (G). This arrangement makes it easier to connect sensors, servos, buzzers, and other modules without complex wiring. Digital devices connect to D pins, analogue sensors connect to A pins, and I<sup>2</sup>C devices such as OLED displays or current sensors connect to A4 (SDA) and A5 (SCL). The board does not change the functionality of the Arduino Nano; it only organises connections neatly and reduces wiring errors. Fig. 4 shows the Arduino Nano mounted on the expansion board.

## Construction and testing

First, upload the source code to the Arduino Nano using the Arduino IDE. After the program uploads successfully, mount the Arduino Nano onto the expansion board carefully and ensure correct alignment of the pins.